



A world powered
by **clean energy**



COMPANY NAME:
InENTec Columbia Ridge LLC

END USE LOCATION:
Arlington, Oregon

QUOTATION #:
H16-1061 - REV 06



DATED:
January 17th, 2019

January 17th 2019

Attn: Jeff Surma
InEnTec Columbia Ridge LLC
Phone: +1 509 946 5700
jesurma@InEnTec.com

Xebec Quotation Number: H16-1061 rev 06

Dear Jeff,

We thank you for your interest in Xebec's PSA technology. For the site condition listed below in section 1.1, we are pleased to provide you with our **firm** proposal as follows:

- One (1) twin stage Xebec PSA30H-G4 / 12HG2-250A-120 system for a fixed prices as outlined in our price section below and in accordance to your syngas specification input detailed in section 1 below.

The proven advantages of the Xebec twin stage G4/G2 PSA system are:

- **Compact System** – Fast PSA cycle that allows adsorber vessels to be 5-15 times smaller than the conventional PSA.
- **Skid Mounted Unit, Pre-loaded with Adsorbents** – Unlike conventional PSAs, field-erection and adsorbent loading is not required for installation.
- **Highly Reliable Rotary Valves (typically <1 CPM)** – Long seal wear life, with 5-Year intervals between servicing.
- **Advanced PSA Nine Adsorber Bed Design in each stage** – Triple equalization PSA cycle, mechanically set to ensure the optimal cycle design.
- **Single Input for Rotary Valve and PSA Cycle Speed**– Simple to operate and requires minimal signals.
- **Optimal Process Cycle Design for Opening & Closing of Ports** – Gradual gas flow changes resulting in very long adsorbent life.

We trust that we have interpreted your requirements correctly and look forward to working with you on this project.

Yours sincerely,

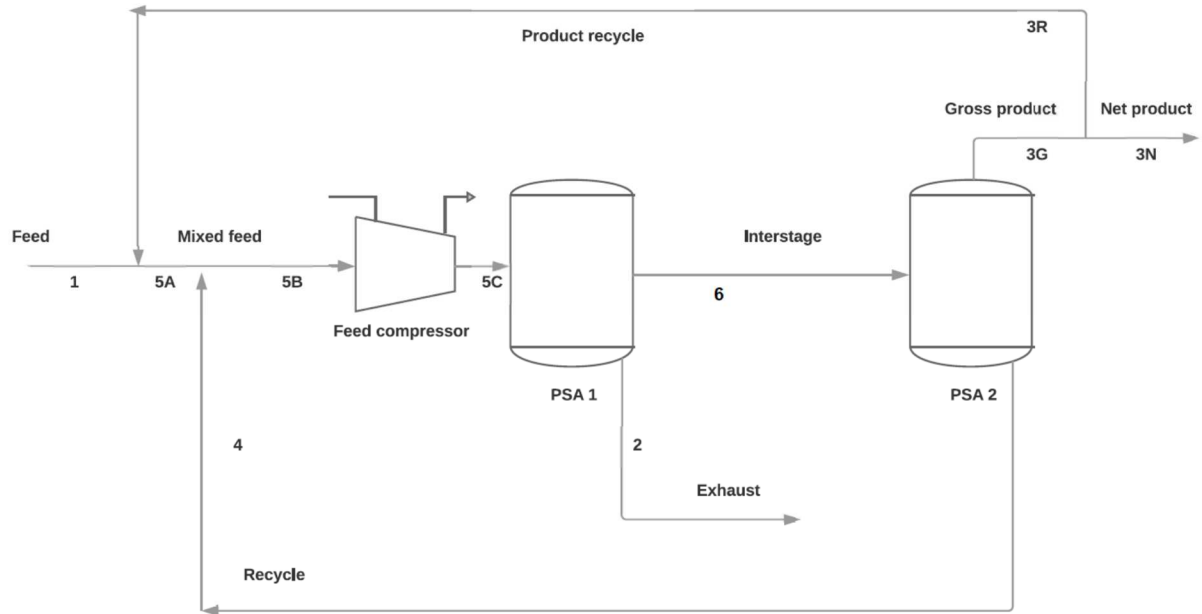
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SECTION 1: SYSTEM COMPONENTS

1.1 DESIGN BASIS

The equipment provided by Xebec pursuant to this proposal will produce on-spec hydrogen (Stream 3 in the diagram below) from purchaser's syngas (Stream 1 in the diagram below) at the recovery rates specified below when processing syngas that meets the inlet syngas specification below:



Flow case HIGH

% H2 Vol	> 99.999
% H2 Vol recovery	79.6% Expected
	70.6% Guaranteed

Estimated pipesizes Parameters	NPS/OD Unit	2" NPS	2" NPS	1.5" NPS	3"/1" NPS	1.5" NPS	6" NPS
		Raw Feed	Mixed Feed	PSA Interstage	Recycle	Product	Exhaust
Flow rate	NCMH	1869	1974	551	105	446	1423
	SCFM	1100	1162	325	62	263	837
Pressure	BARG	13.8	13.8	13.1	0.1/14.5	12.4	0.1
	PSIG	> 200	200	190	2/210	180	2
Temperature	°C	40	40	40	40	40	40
	°F	104	104	104	104	104	104
Composition	% H2 Vol	> 30.0	32.4	95.2	75.0	99.99	8.0
	% CO2 Vol	< 23.0	22.2	1.6	8.2	< 1 PPMV	30.2
	% CO Vol	< 40.0	38.6	2.7	14.3	< 0.2 PPMV	52.6
	% CH4 Vol	< 2.0	1.9	0.1	0.7	< 2 PPMV	2.6
	% N2 Vol	< 5.0	0.0	0.0	0.0	0.0	0.0
	% O2 Vol	< 0.0	0.0	0.0	0.0	0.0	0.0
	% H2O RH	100.0	< 100.0	0.0	0.0	0.0	18.0
	H2O dew temp C	40	35	n.d	n.d	n.d	5.0
	% Total Vol	100.00	100.00	100.00	100.00	100.00	100.00

* Guaranteed

Flow case LOW:

% H2 Vol	> 99.999
% H2 Vol recovery	81.9% Expected
	72.9% Guaranteed

Estimated pipesizes Parameters	NPS/OD Unit	1.5" NPS	1.5" NPS	1" NPS	0.5" NPS	1" NPS	6" NPS
		Raw	Mixed	Interstage	Recycle/Exh2	Product	Exhaust
Flow rate	NCMH	1019	1078	309	59	250	769
	SCFM	600	635	182	35	147	453
Pressure	BARG	13.8	13.8	13.1	0.1/14.5	12.4	0.1
	PSIG	> 200	200	190	2/210	180	2
Temperature	°C	40	40	40	40	40	40
	°F	104	104	104	104	104	104
Composition	% H2 Vol dry	> 30.0	32.5	95.2	75.0	99.99	7.2
	% CO2 Vol dry	< 30.0	28.9	2.0	10.7	< 1 PPMV	39.8
	% CO Vol dry	< 30.0	28.9	2.0	10.7	< 0.2 PPMV	39.8
	% CH4 Vol dry	< 2.0	1.9	0.1	0.7	< 2 PPMV	2.7
	% N2 Vol dry	< 0.0	0.0	0.0	0.0	0.0	0.0
	% O2 Vol dry	< 8.0	7.7	0.6	2.9	< 10 PPMV	10.6
	% H2O RH	< 30.0	32.5	95.2	75.0	99.99	7.2
	H2O dew temp C	40	35	n.d	n.d	n.d	5.0
	% Total Vol	100.00	100.00	100.00	100.00	100.00	100.00

* Guaranteed

PROCESS CONDITIONS

- Exhaust & Product Temperatures: $\pm 10^{\circ}\text{C}$ from Feed Temperature expected but highly dependent on ambient Temperature.
- Water: Design basis of 100% relative humidity (RH) with zero liquid water in feed gas to the PSA.
- Flow units of NCMH are at 0°C and 101.3 KPa(a). Flow units of SCFM are at 32°F and 14.7 PSIA.
- Zero particulate matter over 0.5 micron in size in the feed stream.
- Guaranteed conditions given are at the nominal conditions stated above.
- Increase in feed flowrate or contaminants of up to 10% the stated values above can be handled by increasing PSA valve speed by 10% above rated speed. In this case a 0.5% drop in overall recovery rate is to be expected.

STANDARD FEATURES & SPECIFICATIONS

- Advanced 9-Bed Fast-Cycle PSA Process with Three Equalizations (each PSA)
- Patent-Protected, Multi-Port Rotary Control Valves for Robust & Reliable Operation
- Compact, Modular, & Skid-mounted Design with Pre-loaded Adsorbents for Quick & Simple Installations in Space-Restricted Layouts
- Xebec standard painting and finishes used throughout the PSA as required for industrial environment
- Minimal Start-Up Time to Full Load Conditions
- Adjustable Cycle Speed that Maximizes Performance with Changing Process Parameters

	MINIMUM	MAXIMUM
Design pressure	0.0 Barg (0 Psig)	17.2 Barg (250 Psig)
Operating Ambient Temperature	-29°C (-20°F)	40°C (104°F)
Process Operating Temperature:	4°C (40°F)	60°C (140°F)
Control Panel Operating Temperature:	0°C (32°F)	40°C (104°F)
Storage Temperature:	-20°C (-4°F)	48°C (118°F)

ELECTRICAL

	Classification	ELECTRICAL POWER SUPPLY
Variable Speed Drive (VSD) located in Control Panel* (CP1):	Non hazardous area	460/575 VAC 3Ph 60 Hz OR 120 VAC 1 Ph for PSA motor, 24 VDC for controls and instruments
UL Certified Motor:	Class I div.2 group B or D	Power supply from CP1
Junction Box - Prox. Switch (JB1):	Class I div.2 group B or D	24 VDC
Proximity Switch:	Class I div.2 group B or D	Power supply from customer's IS barrier through JB1
Variable Speed Drive (VSD) located in Control Panel* (CP1):	Non hazardous area	460/575 VAC 3Ph 60 Hz OR 120 VAC 1 Ph for PSA motor, 24 VDC for controls and instruments
UL Certified Motor:	Class I div.2 group B or D	Power supply from CP1

Protection Rating:	NEMA 4 steel panel, internal components suitable for class 1 div 2 group B (methane) or D (hydrogen)
North American standards:	CEC C22.1, NEC, NFPA 70 - National Electric Code, NFPA 70, C22.1, NFPA70 (HazLoc-NA) Ex protection: HazLoc-NA, Class I (gas), Division 2 Group B (H2) or D such as AExia/b, AExd, AExe and others as required
Standard control panel:	NEMA 4 steel panel
Power Consumption:	2 Amps (10 Amps max.)
Vessels:	ASME Sec. VIII, Div. 1, Manufacturer's affidavit for North America
Protection Rating:	NEMA 4 steel panel, internal components suitable for class 1 div 2 group B (methane) or D (hydrogen)

STANDARD FEATURES & SPECIFICATIONS (cont.)	GENERAL ARRANGEMENT	
	Preliminary Plot Plan:	10.0m x 3.4m x 4.1m high (32.8' x 11.3' x 13.3' high)
	Required Foundation:	Concrete pad to be provided by Purchaser
	Estimated weight:	35,000 Kg (77,770 Lbs)

Note: All dimensions, weights and sizes and piping specifications shown here are for information purposes only and may vary depending on model capacity and options.

1.3 INCLUDED EQUIPMENT & AVAILABLE OPTIONS

Selected options in this proposal are given in Section 2: Financial Proposal. Detail descriptions of the included equipment and available options are provided in this section.

1.3.1 Included Equipment:

INCLUDED EQUIPMENT	AUTOMATIC CONTROL PACKAGE(STANDARD)
	This package includes all equipment necessary for safe, unattended, auto-control and turndown operation of the PSA when connected to the Purchaser's PLC with Purchaser's PLC programming approved by Xebec.

A detailed description of the included equipment is given below:

INCLUDED EQUIPMENT		INCLUDED IN PROPOSAL
G4/G2 PSA GENERAL		
Two set of Proprietary and Patent-Protected, Multi-Port Rotary Control Valves		•
Eighteen Adsorbent Beds (Painted RAL 9016 "White") and first load of adsorbent media		•
Common PSV vent header		•
Motor, Gearbox, Shaft, Shaft-Guard, and proximity switch for indicating valve rotation & Adsorbent Bed Protection Trip		•
Variable speed drives and hardwire logic control panel for installation in non-hazardous area		•
Class 1 div 2 skid mounted control panel with Ethernet connection	OPTIONAL	
Piping/tubing connections and flanges to Battery Limits		•
Interconnecting Process Tubing and Piping for Rest-of-Plant		•
(Nine) Pressure Relief Valves with Vent Caps for local venting to atmosphere for each of the two PSA systems		•
Terminal Boxes		•
Skid-mounted Frame		•
First Bed Pressure Sample Port with Pressure Gauge, transmitter and Isolation Valve		•
FEED		
ON/OFF Isolation Valve rated as per Area Classification		•
Pressure Gauge with 2 (1/2" NPT) Sample Ports (or for N2 Purge) and Isolation Ball Valve		•
Coalescing Filter (0.01 micron)		•
Auto-float Liquid Drain Trap including 3-way T-Port Isolation Valve		•
Solenoid Drain Valve with Isolation Valve rated as per Area Classification		•
PRODUCT		
ON/OFF Isolation Valve rated as per Area Classification		•
Particulate Filter (1 micron)		•
Pressure Gauge with 2 (1/2" NPT) Sample Port and Isolation Valve		•
In-Line Spring-Assisted Check Valve		•
Product Direct-Operated Back-Pressure Valve (before discharge to battery limits)		•
PSA product purge line with back pressure regulator, pressure gauge and manual isolation valve		•
EXHAUST		
Pressure Gauge with 2 (1/2" NPT) Sample Port and Isolation Valve		•
Manual Low Point Drain Valve		•
DOCUMENTATION		
One electronic copy and one paper copy of the Installation, Operating, & Maintenance Manual (in English)		•

1.4 MAINTENANCE & RECOMMENDED SPARE PARTS

Prior to delivery, the G4/G2 PSA will be pressurized with nitrogen or air to ensure containment, checked for valve rotation and synchronization, tested for mechanical operation, and tested for other electrical connections.

A summary of the recommended spare parts and maintenance intervals are provided below as a general guideline. A detailed inspection and preventive maintenance chart is provided in the Installation, Operation, and Maintenance manual.

MAINTENANCE	PERIODIC INSPECTION (10 minutes / inspection) Periodic inspection is recommended for preventative maintenance. Motor, gearbox, and valves are inspected for irregular noises, leaks, and obvious degradation of connections (e.g. – grounding, electrical, brackets). Coalescing filter is inspected and condensate is drained at various collection points. Inspection intervals will vary from every 2-3 days to weeks depending on the severity of service and environmental conditions.
	2.5 YEAR SERVICE (2 day / 2.5 years) 2.5 Year Service of the complete system is recommended for optimal performance tuning and overall system adjustment. Internal components of the rotary valve are inspected and parts are replaced as required. Instruments and equipment are tested and checked for accuracy. Filter element is replaced as required. All motor, gearbox, and mechanical couplings are lubricated and/or oil changed as required. Recommended 2.5 Year Service is to be done by Xebec approved personnel.
	5 YEAR SERVICE (2 day / 5 years) Various internal components of the rotary valve are replaced, cleaned, and corrected for alignment. The 5-year service is to be done by Xebec approved personnel only.
SPARE PARTS	ROTARY VALVE SPARE PART KIT (recommended - price available upon request) Valve Seals and Service Kit
	GENERAL SPARE PART KIT (recommended - price available upon request) Variable Speed Drive (VSD); Filter Elements for Xebec supplied filters; Two Regulator Service Kits (for back-pressure valves); Solenoid Valves (Isolation valve)

1.5 ENVIRONMENTAL ASPECTS

- The unit will not produce any gaseous or liquid effluent other than the Product and Exhaust gases (and moisture condensate from feed gas in some installations).
- Solid effluents will consist of used filter cartridges only. The Purchaser is to dispose effluents in satisfactory manner and in compliance with all applicable laws including environmental laws. Purchaser covenants and agrees with Xebec to indemnify Xebec against all liabilities, claims, demands, actions, causes of action, damages, losses, costs or expenses (including legal fees and disbursements) suffered or incurred by Xebec by reason of or arising out of Purchaser's breach of this covenant.

1.6 PURCHASER SCOPE OF SUPPLY

PURCHASER SCOPE OF SUPPLY	EQUIPMENT • Level concrete pad for foundation and anchor bolting. • Piping and fittings for connection to PSA. • PSA exhaust collection system shall not inhibit or restrict PSA exhaust gas flowrate, and shall limit PSA exhaust gas pressure to no more than 2 psig as measured at PSA skid edge. • The syngas feed system shall include overpressure protection for feed conditions above 250 psig. • Control system interface/signals as required by Xebec PSA control logic. • Provision for gradual feed pressurization used on initial start-up. • Provisions for loading PSA media on site (scaffolding, cranes, etc). • Utilities Hook up, nitrogen purge (for maintenance), and instrument air (for electro-pneumatic product control valve option) by Purchaser. • Installation of the panel CP1 in a non-hazardous (non classified) area, with A/C or good air circulation, electrical wiring of CP1 to his controller and power source, electrical wiring of CP1 to the PSA motor. CP1 contains mainly the VFD, power supply and pre-wired connection terminals on which the purchaser should connect the remote signal of his controller (PLC, DCS or else). The panel may contain other items depending on the selected options. The purchaser is responsible to provide the power source and protection to the panel. The panel should be located at no more than 50m from the PSA motor. This point is not required in case panel purged option is purchased with the PSA. • Electrical wiring of Junction boxes (JB1 and JB2) to purchaser's controller (PLC or DCS), JB1 and JB2 are installed on the PSA skid and pre-wired (by Xebec) to the devices of the PSA (SV, proximity, etc) This point is not required in case panel purged option is purchased with the PSA. • Site preparation, off-loading and installation. • Any PSA equipment integration, site evaluation, process hazard studies, risk assessment studies (e.g. – FMEA, HAZOP) TESTING • Pressure testing of connected system. • Correct rotation of motor. • Correct connection of electrical components. • Control System and I/O checkouts. • Correct programming of control logics (supplied by Xebec) including fail-safe protection trip. ADMINISTRATIVE • Permits, licenses and approvals. • Fire protection systems. • Any item not detailed in this document. • Consumables (except for initial beds and any replacement beds required to meet the Process Performance Test).
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SECTION 2: FINANCIAL PROPOSAL

2.1 PRICE SCHEDULE

2.1.1 Xebec Scope of Supply & Price:

SCOPE OF SUPPLY & PRICE		PRICE (\$ USD)
STANDARD FEATURES & SPECIFICATIONS		
One (1) PSA30H-G4 / 12H-G2-250A-120		\$ 800,400
2 x 100% polishing columns on product line Including particulate filter and manual switchover valves and 2 x 16 cuft available bed volume		\$ 36,700
Heat tracing of wet lines and panel heaters for low ambient temperatures		\$ 5,700
Pipesize increase to handle additional flow rated for shifted gas composition of 2400 NCMH raw PSA feed.		\$ 34,000
Seismic and wind load calculations		\$ 6,500
DDU delivery and insurance to Oregon site, including cargo insurance		\$ 19,750 Estimated See shipping terms below
Upgrade to skid mounted PLC control panel suitable for hazardous area with Ethernet IP / Modbus TCP communication protocol/RJ485 port for single point interface communication and power interface with client control/MCC system.		\$ 25,000
On site PSA media loading, commissioning and performance test attendance (3 days on site)		\$ 13,250
Second field visit in case performance test is delayed beyond commissioning date (1 to 2 days on site)		\$ 7,650
Common PSV vent manifold piped to PSA exhaust line		Included
DOCUMENTATION		
Installation, Operating, & Maintenance Manuals	One electronic and one hard copy in English	Included
	Additional Electronic Copies	\$100 ea.
	Additional printed Copies	\$200 ea.
SPARE PARTS (as defined Section 1.3)		
Rotary Valve Spare Part Kit		Avail. Upon request
General Spare Part Kit		Avail. Upon request
DELIVERY AND FREIGHT		
Guaranteed EX WORKS delivery		31 weeks ARO
Xebec Standard Packaging		Included
Shipping Terms		EXW Incoterm. Price option for DDU delivery to site
Insurance		By Purchaser

Taxes and duties are extra. Minimum order is 1 unit(s).
This proposal is valid for a period of one (1) month.

The unit will be ready to ship at the delivery time above, provided the purchaser has approved all Xebec supplied drawings and equipment within one week from receipt from Xebec. Customer approval of Xebec drawings and documents are not required for standard orders.

Although every effort is made at the quotation stage to include the most acceptable materials and latest designs for our products, Xebec reserves the right to alter, modify, or change these systems as necessary to conform to its latest designs at the time of order. Any changes will be conveyed to the Purchaser. Should the Purchaser require any modification, alteration, or change, supplier reserves the right to adjust the proposal prices or deliveries accordingly.

2.1.2 Product Support:

PRODUCT SUPPORT	COMMISSIONING BY XEBEC (quoted at additional price)
	In accordance to our latest service rate plus mobilization charges including transport, accommodation, based on a 7.5 hour day, 5 day week, Monday to Friday. Additional hours and weekends/holidays will be charged at premium rate.

2.2 PROJECT SCHEDULE

2.2.1 Twin stage PSA project schedule:

Deadline (End of Week)	Task
	Issuance of Purchase Order
Week 2 (i.e., 14 days after PO)	Submittal to Purchaser of progress update
Week 4	Submittal to Purchaser of progress update including long lead item purchase orders
Week 6	Submittal to Purchaser of progress update
Week 8	Submittal to Purchaser of progress update including medium lead item purchase orders
Week 10	Submittal to Purchaser of progress update and GA, P&ID, electrical schematics, other drawings with approval / comment-for-resolution
Week 12	Submittal to Purchaser of progress update including short lead item purchase orders
Week 14	Submittal to Purchaser of progress update
Week 16	Submittal to Purchaser of progress update
Week 18	Submittal to Purchaser of progress update
Week 20	Submittal to Purchaser of progress update
Week 22	Submittal to Purchaser of progress update
Week 24	Submittal to Purchaser of progress update and Notice of

	planned Mechanical Inspection and Operation Test date to coordinate schedules and allow time for travel arrangements
Week 26	Submittal to Purchaser of progress update
Week 28	Submittal to Purchaser of progress update
Week 29	Mechanical Inspection and Operation Test
Week 31	Ready to Ship

2.2.1 G2-3 repair schedule:

Deadline (End of week)	Task
	Issuance of Purchase Order
Week 1 (i.e., 7 days after PO)	Pickup and shipment of existing G2 unit from Purchaser
Week 2	Receipt of existing G2 at Xebec
Week 4	Submittal to Purchaser of progress update and notice of intended Mechanical Inspection and Operation Test date to coordinate schedules and allow time for travel arrangements
Week 6	Mechanical Inspection and Operation Test and ready to ship existing G2 unit from Xebec to Purchaser
Week 7	Receipt of existing G2 at Purchaser facility and arrival of Xebec technician for system installation and commissioning

2.3 CREDIT TERMS:

Credit is approved for Purchaser to the Terms of Payment (Section 2.4), subject to ongoing review.

2.4 TERMS OF PAYMENT

2.4.1 Payment Schedule:

Payment Due:

40% (forty) of equipment price: Upon receipt of purchase order

20% (twenty) of equipment price: Upon submission of preliminary GA, P&ID, electrical schematics AND achievement of required hydrogen purity on prior G2 PSA upgrading job at Inentec site in Richland, WA (product purity only).

Client will be allowed to cancel the PO related to this new job/proposal and receive a refund of the prior milestone associated to this new job/proposal less a 2.5% restocking fee should the smaller G2 system not achieve stated product purity.

20% (twenty) of equipment price: Upon FAT completion

10% (ten) of equipment price: Upon readiness to ship

10% (ten) of equipment price: Upon successful completion of performance test for the twin stage PSA

Note: In order for the performance warranty to be valid, the equipment must be started by a Xebec authorized service technician.

2.4.2 Payment Details:

Late payment charges are 2% **outstanding amount** per month. All payments are to be considered non-refundable deposits. In the event that payment is not received within 30 days of the above terms, Xebec has the option **to suspend performance of this agreement and toll the delivery date.**

2.5 CONDITIONS OF CONTRACT:

2.5.1 Conditions

Any purchase order issued by Purchaser in respect of this Proposal will be subject to acceptance by Xebec and the terms and conditions of this Proposal, and, in the event of any inconsistency between this Proposal and the purchase order, this Proposal will govern unless Xebec expressly consents in writing to a term of the purchase order overriding this Proposal. If any purchase order introduces a term or condition not covered by this Proposal, then such term or condition shall not bind Xebec unless Xebec expressly consents in writing to such term or condition.

2.5.2 Standard Conditions:

Xebec Adsorption's "General Terms and Conditions of Sale" contained in section 3 below shall apply.

2.5.3 Late Delivery Penalty:

Xebec will provide purchaser with bi-weekly updates on progress against the schedule defined in this proposal. Progress updates shall include verification of equipment orders and receipts (redacted POs, shipping invoices, receipt inspections, etc.) and/or onsite inspections and verification by Purchaser. A penalty of 1.25% of the equipment purchase price (not to exceed 5% total) shall be withheld for every week delayed past the agreed upon schedule. Penalties shall be withheld from the upcoming payment due Xebec. Xebec can recoup the penalties incurred if it recovers schedule to eliminate delays. Xebec can earn a bonus of 1% per week up to 5 weeks for early delivery.

2.4.4 Warranty Repair of Previously Purchased G2 Unit:

Xebec agrees to repair/modify the G2 system previously purchased by InEnTec Inc. (Model# PSA3H-G2-330A-480, Job# 30784) at its cost so to achieve the guaranteed product purity specifications with a feed flow rate range between 7 to 32 scfm and an expected (non guaranteed) recovery rate between 27% to 55% depending on feed gas flow and composition.

InEnTec will ship the G2 to Xebec at Xebec's cost and Xebec will make repairs at its factory, and ship the unit back to Richland, WA (at Xebec's cost) for InEnTec to confirm that the unit meets the performance specification while processing InEnTec's syngas.

Any onsite work in Richland for re-commissioning and startup will be at Xebec's cost.

Client will be allowed to cancel the PO related to this proposal and receive full refund of prior milestone payment associated to this new job/current proposal should the smaller G2 system not achieve stated product purity.

SECTION 3: GENERAL TERMS AND CONDITIONS OF SALE

Shall constitute a part of Xebec's standard sale agreement

3.0 Definitions:

3.0.1 "Goods":

All the agreed upon items and associated ancillary equipment that constitutes due fulfillment of this project.

3.0.2 "Supply":

To organize, design, manufacture, purchase, expedite, assemble, and test all the "Goods" F.O.B., at Xebec's premises, in either fully assembled, partially assembled or fully dismantled form.

3.0.3 "Transport":

To organize and provide suitable transport for the delivery of all the "Goods" from Xebec's premises to the Purchaser's site.

3.0.4 "Inspection":

To witness, by a Purchaser's authorized representative, suitable "factory testing" at Xebec's premises of all "Goods".

3.0.5 "Installation":

To set out, place in position, line up, and connect up all "Goods" to other installations or equipment, and to plant services provided, using the Purchaser's installation team.

3.0.6 "Commissioning":

To provide a Xebec certified technician, or other such suitably qualified person, for the tuning of mechanical and electrical components and putting the "Goods" in operational mode to allow commercial and/or testing use.

3.1 Prices:

Unless otherwise specified by Xebec, Xebec's price for the Goods shall remain in effect for two months after the date of Xebec's quotation or acceptance of the order for the Goods, whichever is delivered first, provided an unconditional, complete authorization for the immediate manufacture and shipment of Goods is received and accepted by Xebec within such time period.

3.2 Delivery and Documentation:

Unless otherwise specified in this proposal, the Purchaser shall pay transportation and handling expenses. Delivery to carrier shall constitute passage of risk of loss to the Purchaser irrespective of arrangements for transportation or insurance charges.

3.3 Shortage:

The Purchaser must inspect Goods promptly upon receipt and must submit any claim for shortage within thirty days after receipt or any such claim shall be waived.

3.4 Cancellation by Purchaser:

If Purchaser cancels this agreement or refuses delivery of the Goods, the Purchaser shall be liable for and shall pay to Xebec all expenses incurred or committed to by Xebec prior to cancellation and Xebec's profit to the same extent as if the Purchaser had accepted delivery of the finished Goods.

3.5 Force Majeure:

Xebec shall not be liable for delays in performance or for non-performance due to acts of God, war, riot, fire, labor **strike**, explosion, accident, storm, flood, earthquake, compliance with governmental requests, laws, regulations, orders or actions, or unforeseen circumstances or causes beyond Xebec's reasonable control.

3.6 Operational and Process Performance Testing

Testing will be performed in the following steps:

3.6.1 Mechanical Inspection and Operational Test

(a) A Mechanical Checkout and Operational Test will be performed on the Goods at Xebec's factory prior to shipment.

- (b) No fluids processing will be performed during this step.
- (c) Purchaser representatives will be invited to see this test. Their travel and living expenses will be at the Purchaser's charge.
- (d) The test will be deemed successful if Xebec's Quality Department ensures all components of the Goods are present, properly installed and operational according to the Scope of Supply in Sections 1 and 2.

3.6.2 Process Performance Test

(a) A Process Performance Test will be performed at Purchaser's site under Xebec representative's supervision following Goods installation, systems checkout and start-up to prove achievement of the guaranteed performance criteria.

(b) Purchaser will give fifteen (15) days of advance notice to the Xebec representative of the Process Performance Test starting day. Such test starting date must be within **one hundred and twenty (120)** days after delivery of the Goods. Xebec will send qualified personnel to assist with the test. If the Purchaser fails to make the system available for Process Performance Test during the specified time period, the system will be deemed successfully complete and accepted by the Purchaser. The Purchaser will take over the Scope of Supply and the Purchaser will release Xebec from Process Guarantee.

(c) Purchaser must provide all specified utilities and site requirements. Fluids, manpower, energy and analysis will be at Purchaser's charge. Analysis of the gas purity will be performed at a mutually acceptable test facility. Any flow measurement, purity analysis or other equipment required to accurately determine process performance will be at Purchaser's charge. The performance test will be subject to the accuracy limitations of those instruments. The Purchaser will provide calibration certificates and accuracy specifications for the instruments required, to the satisfaction of Xebec, prior to the Performance Test.

(d) A **forty-eight (48)** hour test period of continuous operation test will be performed and all operating characteristics will be recorded and presented in report form to Purchaser. This test will be deemed successful if the process guarantee conditions specified in section 1 are met.

(e) If conditions exist at the site beyond parties' control that prevent the successful performance of the test, parties will mutually agree to a different test schedule.

(f) Upon completion of a successful Process Performance Test, both parties will sign a certificate of successful performance of the Process Performance Test; and the Purchaser will take over the Scope of Supply **and the Purchaser will release Xebec from Process Guarantee.**

(g) In case the Goods fail to reach the guaranteed performance by reasons for which Xebec is responsible, Xebec's sole responsibility shall be, at its option either, (i) commence actions to carry out, on a best effort basis, at his own costs and within 1 (one) month after the Process Performance Test or such other time as the parties shall agree to, all remedial works that are necessary to achieve the guaranteed performance in a further Process Performance Test or (ii) refund the purchase price paid by Purchaser.

3.7 Warranty:

Xebec warrants the Goods against defects in workmanship or materials under normal use for twelve months from date of start-up or eighteen months from date of shipment, whichever is earlier. All replacements or repairs necessitated by inadequate preventive maintenance, or by normal wear and usage, or by fault of the Purchaser, or by unsuitable power sources or by attack or deterioration under unsuitable environmental conditions, or by abuse, accident, alteration, misuse, improper installation, modification, repair, storage or handling, or any other cause not the fault of Xebec ("Non-warranty Repairs") are not covered by this Limited Warranty, and shall be at Purchaser's expense. All costs of dismantling, reinstallation and freight and the time and expenses of Xebec's personnel for site travel and non-warranty repairs shall be borne by Purchaser unless accepted in writing by Xebec. Xebec's sole responsibility shall be, at its option, to repair or replace and install free of charge any parts or equipment found to be defective, up to a maximum of 50% of the purchase price, or refund the full purchase price paid by the Purchaser. Goods repaired and parts replaced during the warranty period shall be in warranty for the remainder of the original warranty period or ninety (90) days, whichever is longer. THE FOREGOING WARRANTIES AND REMEDIES ARE EXCLUSIVE AND EXPRESSLY IN LIEU OF ALL OTHER WARRANTIES EXPRESSED OR IMPLIED; AND XEBEC MAKES NO REPRESENTATION OR WARRANTY, EXPRESS OR IMPLIED, AS TO THE FITNESS OF A PARTICULAR PURPOSE OR MERCHANTABILITY OF THE GOODS.

3.7.1 Equipment Access

After providing reasonable notice to the Purchaser, Xebec personnel must have access to the purchaser site to collect performance data and samples during the warranty period.

3.8 Limitation of Remedy and Liability:

XEBEC SHALL NOT BE LIABLE FOR DAMAGES CAUSED BY DELAY OR PERFORMANCE. THE SOLE AND EXCLUSIVE REMEDY FOR BREACH OF WARRANTY HEREUNDER SHALL BE LIMITED TO REPAIR, CORRECTION OR REPLACEMENT UNDER THE LIMITED WARRANTY CLAUSE IN SECTION 3.7. IN NO EVENT, REGARDLESS OF THE FORM OF THE CLAIM OR CAUSE OF ACTION (WHETHER BASED IN CONTRACT, INFRINGEMENT, NEGLIGENCE, STRICT LIABILITY, OTHER TORT OR OTHERWISE), SHALL XEBEC'S LIABILITY TO PURCHASER EXTEND TO INCLUDE INCIDENTAL, CONSEQUENTIAL OR PUNITIVE DAMAGES. THE TERM "CONSEQUENTIAL DAMAGES" SHALL INCLUDE,

BUT NOT BE LIMITED TO, LOSS OF ANTICIPATED PROFITS, LOSS OF USE, LOSS OF REVENUE, AND COST OF CAPITAL.

3.9 Installation:

The Purchaser shall be responsible for receiving, storing, installing, starting up, and maintaining all Goods. Xebec shall provide a quotation for services to assist the Purchaser in these functions if requested. In order for the warranty to be valid, commissioning and start-up are to be performed by a Xebec certified technician.

3.10 Product Support:

Xebec's policy is that functionally interchangeable replacement parts will be available during the time a standard product is offered for sale. In addition, either repair capability, or functionally equivalent parts will be available for at least five years from the last date of product availability. No guarantee of parts availability, repair capability, or functionally equivalent goods is offered for goods manufactured by others and supplied with Xebec's Goods.

3.11 Taxes and Duties:

Xebec's prices do not include amounts for any duties or sales, use, excise, or similar taxes. The Purchaser shall pay, in addition to Xebec's prices, the amount of any present or future such tax and duty applicable to the sale or use of the Goods or, in lieu of such payment, provide Xebec with a tax exemption certificate and/or duty clearance acceptable to the appropriate authorities. The Purchaser covenants and agrees with Xebec to indemnify Xebec against all liabilities, claims, demands, actions, causes of action, damages, losses, costs or expenses (including legal fees and disbursements) suffered or incurred by Xebec by reason of or arising out of Purchaser's breach of this covenant.

3.12 Insurance:

The Purchaser shall provide and maintain insurance on the Goods in an amount not less than their full insurable value, with loss payable to Xebec and the Purchaser, as their interests may appear, from the date that risk of loss passes to the Purchaser until such time as title passes to the Purchaser and the Purchaser pays the full purchase price to Xebec.

3.13 Labeling:

Each of the Goods utilized by the Purchaser shall have a label affixed thereto identifying it as manufactured or developed by Xebec and containing such proprietary, copyright, patent, trademark, design right, trade secret and all other proprietary rights, legends and alpha-numeric codes as Xebec considers appropriate. The Purchaser shall ensure that such label remains affixed to such Goods, shall replace such label if it is destroyed, removed or becomes unreadable and in no event shall the Purchaser claim that the Goods or any component thereof has been developed by the Purchaser.

3.14 DELETED

3.15 DELETED

3.16 Storage:

The Purchaser agrees to pay Xebec a storage charge of 1% of the total purchase price per month, or prorated portion thereof for a part of a month, for Goods not authorized for shipment by the Purchaser within fourteen (14) days following either the specified delivery date or the date the Purchaser is notified that same are ready for shipment, whichever is later.

3.17 Title

(a) Purchaser agrees to grant to Xebec such security interests in the Goods and to execute such other documents as may be desirable for Xebec to obtain a perfected first ranking charge over the Goods as security for the payment of all monies owing to Xebec by the Purchaser and performance of all obligations of the Purchaser under this agreement.

(b) It will be a default if the Purchaser fails to make any payment due under this Agreement or any other agreement between the Purchaser and Xebec when due.

(c) Until Xebec has been paid in full for the Goods in accordance with this Agreement, title to the Goods shall remain exclusively in Xebec.

(d) Purchaser acknowledges receipt of a copy of this Agreement and waives all rights to receive from Xebec a copy of any financing statement, financing change statement, or verification statement filed at any time at the Personal Property Registry or other similar registry in respect of this Agreement.

(e) In the event of default of this Agreement or in the payment of any sums due to Xebec as aforesaid, Xebec may exercise any and all remedies afforded to secured parties by Part 5 of the *Personal Property Security Act* or other similar statutes, including seizure of the Goods.

3.18 Proprietary Rights

(a) Xebec shall defend at its own expense any legal proceeding brought against the Purchaser and pay those damages finally awarded against the Purchaser in such proceeding, to the extent that it is based on a claim that the Goods in the form delivered

to the Purchaser constitutes an infringement of any valid patent trademark, copyright, trade secret or other third-party proprietary right held under the law of the country in which the Site is located.

The Purchaser shall notify Xebec promptly in writing of any claim of infringement.

(b) Xebec shall have no liability or obligation to the Purchaser hereunder with respect to any patent, trademark, copyright, trade secret or other third-party proprietary right infringement or claims thereof based upon (i) unauthorized modifications of the Goods, (ii) any claim of patent, trademark, copyright, trade secret or other third-party proprietary right infringement in which the Purchaser or any affiliate or purchaser of Purchaser has an interest or license.

(c) In the event that any of the Goods is held to constitute infringement or the use thereof is enjoined, Xebec may at its option and at its own expense, as the exclusive remedy to the Purchaser therefor, either, (i) procure for the Purchaser the right to continue using such Goods, or (ii) replace such Goods with non-infringing Goods of equivalent quality and performance or, (iii) modify such Goods so that it is non-infringing, or (iv) accept return of such Goods.

(d) The foregoing states the entire liability of Xebec with respect to infringement of patents, trademarks, copyrights, trade secrets, or other third-party proprietary rights by any Goods delivered under this Agreement.

(e) The sale acceptance of the any Goods to the Purchaser by Xebec shall not confer on the Purchaser a license under any patent, trademark, copyright, trade secret or other proprietary rights of Xebec, and shall not confer on the Purchaser a license to reproduce, redesign, modify, create derivative works from, reverse engineer, disassemble or reverse assemble the Goods.

3.19 General Provisions:

(a) These terms and conditions, and the documents to which they are attached, constitute the entire agreement between the parties regarding the sale and purchase of the Goods and/or services to be performed and no other promises or agreements shall be of any force or effect unless mutually agreed upon in writing.

(b) In the case of contradictory terms or conditions of sale between this document and the attached documents which constitute the entire agreement, the terms or conditions of the attached documents shall take priority.

(c) These terms shall be binding upon and shall accrue to the benefit of the parties hereto and their respective assigns and successors in interest; provided, however, neither party may assign this agreement, or any of its interests herein, without the prior written consent of the other party.

(d) No action, regardless of form, may be brought by either party more than two (2) years after the cause of the action has accrued.

(e) The laws of the Province of Québec, Canada shall govern the validity, interpretation, and performance of this agreement. Canadian Courts shall adjudicate all disputes and all judgments shall be binding and final.

(f) Purchaser represents and warrants that Goods supplied hereunder are not to be used in nuclear applications, including, without limitation, any nuclear power generation facility.